Vector Spaces

<u>Definition (Vector Space)</u>. Fix a field F of scalars. Roughly a vector space V over the field F is an abelian group with a "compatible" scalar multiplication. More precisely, we have

- $\vec{v}, \vec{w} \in V \Rightarrow \vec{v} + \vec{w} \in V$ (closure under addition).
- $c \in F$, $\vec{v} \in V \Rightarrow c\vec{v} \in V$ (closure under scalar mult.)

<u>Axiom 1 (Scalar multiplication association)</u>

$$c, d \in F, \ \vec{v} \in V : c(d\vec{v}) = (cd)\vec{v}$$

<u>Axiom 2 (Distribution)</u>. $c, d \in F$, $\vec{v}, \vec{w} \in V$,

$$c(\vec{v} + \vec{w}) = c\vec{v} + c\vec{w},$$
$$(c+d)\vec{v} = c\vec{v} + d\vec{v}.$$

<u>Axiom 3 (Scalar identity)</u>. $1 \in F$, $\vec{v} \in V$

$$1\vec{v} = \vec{v}$$

Axiom 4 (Group Axioms for $(V, +)$).

$+$ is associative, $\vec{v} + \vec{w} = \vec{w} + \vec{v}$.
Zero vector: $\vec{0} + \vec{v} = \vec{v} = \vec{v} + \vec{0}$
Additive inverses: $\vec{v} + (-\vec{v}) = \vec{0}$.

<u>Example.</u> $V = F^n = \left\{ \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \;\middle|\; a_i \in F \right\}$ is a vector space over F.

$\{ (a_1 \; \cdots \; a_n) \mid a_i \in F \}$ is a vector space over F.

$V = \mathrm{Mat}_{n \times m}(F)$. Operations:
- Addition
- scalar multiplication

Polynomial Spaces: Fix $k \geq 0$. Then

$$V = \mathbb{P}_k(F) = \left\{ \sum_{i=0}^{k} a_i x^i \;\middle|\; a_j \in F \right\}$$

Operations: Let $p(x) = \sum_{i=0}^{k} a_i x^i$, $q(x) = \sum_{i=0}^{k} b_i x^i$, $c \in F$, then

$$p(x) + q(x) = \sum_{i=0}^{k} (a_i + b_i) x^i, \quad cp(x) = \sum_{i=0}^{k} c a_i x^i$$

E.g. $\mathbb{P}_0(F) = \{a \mid a \in F\} = F$.

$\mathbb{P}_1(F) = \{a + bx \mid a, b \in F\}$

$\mathbb{P}_2(F) = \{a + bx + cx^2 \mid a, b, c \in F\}$

Note: $\mathbb{P}_0(F) \subseteq \mathbb{P}_1(F) \subseteq \mathbb{P}_2(F) \subseteq \cdots$

Set $\mathbb{P}(F) = \bigcup_{k \geq 0}^{\infty} \mathbb{P}_k(F)$, then $V = \mathbb{P}(F)$ is also a vector space by adding and scaling polynomials.

Function spaces: For F a field, X a set, define

$$F^X := \{f \mid f : X \to F\}$$

is a vector space over F.

E.g. $\mathbb{R}^{[0,1]} = \{f \mid f : [0,1] \to \mathbb{R}\}$.

Operations on F^X: "pointwise operations".

- $f : X \to F$, $g : X \to F$ $\rightsquigarrow$ $f + g : X \to F$, $(f+g)(x) = f(x) + g(x)$

- $c \in F$, $f : X \to F$ $\rightsquigarrow$ $cf : X \to F$, $(cf)(x) = cf(x)$

- zero vector: $0(x) = 0 \in F$, zero function.

In $\mathbb{R}^{\mathbb{R}} = \{ f \mid f : \mathbb{R} \to \mathbb{R} \}$, we often focus on subsets of functions with nice properties, e.g.

$C(\mathbb{R})$: continuous functions
$C^1(\mathbb{R})$: differentiable functions
$C^2(\mathbb{R})$: twice differentiable functions
$\vdots$
$C^\infty(\mathbb{R})$: smooth functions.

Subspaces

<u>Definition (Subspace)</u>. Let V be a vector space over F. A subset $W \subseteq V$ is called a subspace of V if

- $W \neq \emptyset$ (non-empty)
- W is itself a vector space over F using operations from V, i.e. if $\vec{v}, \vec{w} \in W$, $\vec{v} + \vec{w} \in W$, $\lambda \in F$, $\vec{v} \in W \Rightarrow \lambda \vec{v} \in W$.

<u>Example</u>. $\{\vec{0}\} \subseteq V$ is always a subspace: $\lambda \vec{0} = \vec{0}$, $\vec{0} + \vec{0} = \vec{0}$.

$V \subseteq V$ is a subspace.

<u>Example</u>. $V = \mathbb{R}^3$, $W = \{(x, y, z) \in \mathbb{R}^3 \mid x + 2y - z = 0\}$.

- Nonempty, as $(0,0,0) \in W$.
- Closure: Let $\lambda \in \mathbb{R}$, $(x, y, z) \in W$, then $(\lambda x, \lambda y, \lambda z)$ satisfies

$$\lambda x + 2\lambda y - \lambda z = \lambda(x + 2y - z)$$
$$= \lambda \cdot 0 = 0.$$

Let $(x, y, z), (a, b, c) \in W$,

$$x + 2y - z = 0 \qquad (1)$$
$$a + 2b - c = 0 \qquad (2)$$

$(1) + (2)$: $(x + a) + 2(y + b) - (z + c) = 0$

$\therefore$ Closed.

$\therefore$ W is a subspace.

<u>Example</u>. In $\mathbb{R}^{\mathbb{R}} = \{f \mid f : \mathbb{R} \to \mathbb{R}\}$, we have subspaces

$$\mathbb{R}^{\mathbb{R}} \supseteq C(\mathbb{R}) \supseteq C^1(\mathbb{R}) \supseteq C^2(\mathbb{R}) \supseteq \cdots \supseteq C^\infty(\mathbb{R})$$

Spans of Vectors

<u>Definition (Linear Combination)</u>.

Let X be a subset of vectors in a vector space V. A linear combination of vectors from X is an expression of the form

$$\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \cdots + \lambda_k \vec{v}_k$$

where $\lambda_i \in F$, $\vec{v}_i \in X$.

<u>Definition (span)</u>. Let V be a vector space, and X be a subset of vectors in V. The span of X, denoted $\langle X \rangle$ is defined to be

- If $X = \emptyset$, set $\langle X \rangle = \{\vec{0}\}$
- If $X \neq \emptyset$, then

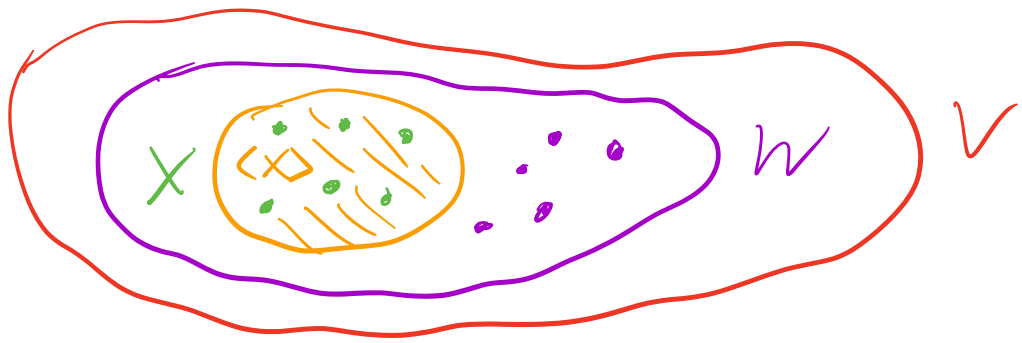
$$\langle X \rangle = \{\lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k \mid \lambda_i \in F, \vec{v}_i \in X\}$$

We call $\langle X \rangle$ the subspace of V spanned by X. If $X = \{\vec{v}_1, \ldots, \vec{v}_k\}$, we often write

$$\langle X \rangle = \langle \vec{v}_1, \ldots, \vec{v}_k \rangle$$

Theorem. Let V be a vector space over F, X a subset of V. If $W \subseteq V$ is a subspace of V which contains X, then $\langle X \rangle \subseteq W$ is a subspace of W.

Intuition: $\langle X \rangle$ is the smallest subspace of V containing X.



If $W = \langle X \rangle$, we say X spans W.

Example. Let $X = \{\vec{e}_1, \vec{e}_2\} \subseteq \mathbb{R}^2$, then X spans $\mathbb{R}^2$:

$$\vec{v} = \begin{pmatrix} a \\ b \end{pmatrix} = a\vec{e}_1 + b\vec{e}_2 \in \langle \vec{e}_1, \vec{e}_2 \rangle$$

$$\Rightarrow \mathbb{R}^2 = \langle \vec{e}_1, \vec{e}_2 \rangle$$

Example. In $V = \mathbb{P}_3(F)$, let

$W = \{p(x) \in \mathbb{P}_3(F) : p(x) \text{ has constant term } a_0 = 0\}$.

W is a subspace spanned by x, x^2, x^3, i.e.

$$W = \langle x, x^2, x^3 \rangle.$$

Row and Column Space of a Matrix

<u>Definition</u>. Let $M \in \mathrm{Mat}_{m \times n}(F)$, $\{R_1, \ldots, R_m\}$ be rows of M, and $\{C_1, \ldots, C_n\}$ be columns of M. The row space of M is given by the span

$$\mathrm{row}(M) = \langle R_1, \ldots, R_m \rangle \subseteq \mathrm{Mat}_{1 \times n}(F) \approx F^n.$$

The column space of M is given by the span

$$\mathrm{col}(M) = \langle C_1, \ldots, C_n \rangle \subseteq \mathrm{Mat}_{m \times 1}(F) \approx F^m$$

<u>Theorem</u>. Let $A, B \in \mathrm{Mat}_{m \times n}(F)$. Then $\mathrm{row}(A) = \mathrm{row}(B)$ iff they are row equivalent (same RREF); $\mathrm{col}(A) = \mathrm{col}(B)$ iff $\mathrm{row}(A^T) = \mathrm{row}(B^T)$.